Study notes

Metric Spaces & Topology

Chapters 0–5

Tuesday 10 November 2:00–7:00 pm session

Definitions, theorems with proofs, worked examples, common mistakes, graded exercises with full solutions, proof templates, a counterexample bank and a glossary of every term needed for the exam.

How to use these notes

Each chapter starts with its **study slot** from the session plan below and a short list of what to focus on in that slot. Inside each chapter:

- **Intuition** paragraphs give the idea in plain English before the formal statement.
- **Definitions** are worded the way you should state them in an exam.
- **Theorems, lemmas, propositions** come with proofs of the results most often examined.
- **Examples** and **Warnings** cover standard cases and common mistakes.
- **Exercises** are tagged [Easy] , [Medium] or [Hard] and each has a full solution. Cover the solution and try the exercise first.

The appendices contain a **proof-template toolkit**, a **counterexample bank**, a **glossary** and the **recap checklist** for the 5:35 slot.

Chapter numbering follows the usual order for a first course in metric spaces. If your lecture notes order the topics differently, go by the section headings.

Session plan: Tuesday 10 November

Time	Task	Where
2:00–2:10	Warm-up: skim the chapter headings	Contents
2:10–2:30	Chapter 0: preliminaries	Chapter 0
2:30–3:05	Chapter 1: metric spaces	Chapter 1
3:05–3:15	Break	
3:15–3:50	Chapter 2: open and closed sets, topologies	Chapter 2
3:50–4:25	Chapter 3: continuity	Chapter 3
4:25–4:40	Longer break	
4:40–5:10	Chapter 4: completeness	Chapter 4
5:10–5:35	Chapter 5: compactness	Chapter 5
5:35–5:45	Recap from memory	Recap checklist
5:45–6:00	Break	
6:00–6:50	Linear systems	Separate notes
6:50–7:00	Wrap-up: what's done, where to start next time	

Notation

(X, d)	a metric space; $x, y, z \in X$ are points
$B(x, r) = \{\, y \in X \;:\; d(x, y) < r \,\}$	open ball of radius $r > 0$
$\overline{B}(x, r) = \{\, y \in X \;:\; d(x, y) \leq r \,\}$	closed ball
$\operatorname{int} A,\ \operatorname{cl} A = \overline{A},\ \partial A$	interior, closure, boundary
$A^c = X \setminus A$	complement
A'	derived set (the limit points of A)
$(x_n)_{n\in\mathbb{N}},\ x_n \to x$	a sequence; convergence
$\mathbb{N} = \{1, 2, 3, \dots\},\ \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	the usual number systems
$C[a, b]$	the continuous functions $[a, b] \to \mathbb{R}$

Contents

How to use these notes **i**

0 Preliminaries **1**

0.1 Logic and quantifiers . . . 1
0.2 Sets . . . 1
0.3 Functions, images and preimages . . . 1
0.4 Countability . . . 2
0.5 Supremum and infimum . . . 2
0.6 Inequalities . . . 2
0.7 Sequences of real numbers . . . 3

1 Metric Spaces **4**

1.1 Definition and first examples . . . 4
1.2 Normed spaces . . . 5
1.3 Further examples and non-examples . . . 6
1.4 Consequences of the axioms . . . 6
1.5 Equivalent metrics . . . 7
1.6 Worked proof: $d/(1+d)$ is a metric . . . 7
1.7 Exercises . . . 7

2 Open and Closed Sets, Convergence, Topological Spaces **9**

2.1 Balls . . . 9
2.2 Open and closed sets . . . 9
2.3 Interior, closure, boundary and limit points . . . 11
2.4 Convergence of sequences . . . 13
2.5 Topological spaces . . . 13
2.6 Subspaces . . . 14
2.7 Exercises . . . 15

3 Continuity and Homeomorphisms **16**

3.1 Four equivalent definitions . . . 16
3.2 Building continuous functions . . . 17
3.3 Uniform continuity and Lipschitz maps . . . 18
3.4 Homeomorphisms . . . 18
3.5 Exercises . . . 19

4 Completeness **21**

4.1 Cauchy sequences 21

4.2 Complete metric spaces 21

4.3 The Contraction Mapping Theorem 23

4.4 Further results 24

4.5 Exercises 24

5 Compactness **26**

5.1 Definitions 26

5.2 The main equivalence 27

5.3 Heine–Borel and its limits 27

5.4 Key theorems 28

5.5 Exercises 29

A Connectedness **31**

B Proof-Template Toolkit **32**

C Counterexample Bank **34**

D Glossary **35**

E Recap Checklist (5:35–5:45) **38**

Chapter 0

Preliminaries

Study slot: 2:10–2:30 (20 min)

Read this quickly: it collects the tools every later chapter uses. Make sure you can apply the preimage identities (Proposition 0.4) and negate quantified statements without hesitation.

0.1 Logic and quantifiers

Almost every definition in this course is a chain of quantifiers $\forall$ and $\exists$. To *negate* such a statement, swap $\forall \leftrightarrow \exists$ and negate the final condition.

Example 0.1 (Negating convergence)**.** The statement $x_n \to x$ means

$$\forall \varepsilon > 0 \;\; \exists N \in \mathbb{N} \;\; \forall n \geq N : \quad d(x_n, x) < \varepsilon.$$

Its negation, $x_n \not\to x$, is

$$\exists \varepsilon > 0 \;\; \forall N \in \mathbb{N} \;\; \exists n \geq N : \quad d(x_n, x) \geq \varepsilon,$$

that is, infinitely many terms stay at least ε away from x.

The proof methods used throughout are: *direct proof*; *contrapositive* (prove $\neg Q \Rightarrow \neg P$ to get $P \Rightarrow Q$); *contradiction*; and *induction*.

0.2 Sets

Proposition 0.2 (De Morgan's laws)**.** *For any family $(A_i)_{i \in I}$ of subsets of X,*

$$X \setminus \bigcup_{i \in I} A_i = \bigcap_{i \in I} (X \setminus A_i), \qquad X \setminus \bigcap_{i \in I} A_i = \bigcup_{i \in I} (X \setminus A_i).$$

These laws turn every statement about open sets into a statement about closed sets, and vice versa. To prove $A = B$, show $A \subseteq B$ and $B \subseteq A$ by taking an arbitrary element.

0.3 Functions, images and preimages

Definition 0.3. Let $f \colon X \to Y$, $A \subseteq X$ and $B \subseteq Y$.

(i) The **image** of A is $f(A) = \{\, f(a) \;:\; a \in A \,\}$.

(ii) The **preimage** of B is $f^{-1}(B) = \{\, x \in X \;:\; f(x) \in B \,\}$. This is defined even if f has no inverse.

(iii) f is **injective** if $f(x) = f(y) \Rightarrow x = y$; **surjective** if $f(X) = Y$; **bijective** if both. Exactly in the bijective case there is an inverse $f^{-1} \colon Y \to X$.

Proposition 0.4 (Preimages preserve set operations)**.** *For any family $(B_i)_{i\in I}$ of subsets of Y and any $B \subseteq Y$,*

$$f^{-1}\Big(\bigcup_i B_i\Big) = \bigcup_i f^{-1}(B_i), \quad f^{-1}\Big(\bigcap_i B_i\Big) = \bigcap_i f^{-1}(B_i), \quad f^{-1}(Y \setminus B) = X \setminus f^{-1}(B).$$

Images only satisfy $f(\bigcup A_i) = \bigcup f(A_i)$ and $f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$, with equality when f is injective. Moreover

$$A \subseteq f^{-1}(f(A)), \qquad f(f^{-1}(B)) \subseteq B.$$

Remark 0.5. This is why continuity is defined through *preimages* ("f^{-1}(open) is open"): preimages respect unions, intersections and complements, and images do not.

0.4 Countability

A set is **countable** if it is finite or in bijection with $\mathbb{N}$. The sets $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ are countable, and so is any countable union of countable sets. $\mathbb{R}$ is **uncountable** (Cantor's diagonal argument).

0.5 Supremum and infimum

Definition 0.6. Let $\emptyset \neq A \subseteq \mathbb{R}$. A number M is an **upper bound** for A if $a \leq M$ for all $a \in A$. The **supremum** $\sup A$ is the least upper bound, and the **infimum** $\inf A$ is the greatest lower bound.

Theorem 0.7 (Completeness axiom of $\mathbb{R}$)**.** *Every non-empty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.*

This is exactly what $\mathbb{Q}$ lacks: $\{\, q \in \mathbb{Q} \,:\, q^2 < 2 \,\}$ has no supremum in $\mathbb{Q}$.

Proposition 0.8 (Approximation property)**.** *If $s = \sup A$, then for every $\varepsilon > 0$ there exists $a \in A$ with $s - \varepsilon < a \leq s$. Similarly, if $t = \inf A$, there exists $a \in A$ with $t \leq a < t + \varepsilon$.*

0.6 Inequalities

For $a, b \in \mathbb{R}$ and $a_i, b_i \in \mathbb{R}$:

Triangle inequality:	$\lvert a+b\rvert \leq \lvert a\rvert + \lvert b\rvert\,,$
Reverse triangle inequality:	$\big\lvert\lvert a\rvert - \lvert b\rvert\big\rvert \leq \lvert a-b\rvert\,,$
Cauchy–Schwarz:	$\Big(\sum_{i=1}^{n} a_i b_i\Big)^2 \leq \Big(\sum_{i=1}^{n} a_i^2\Big)\Big(\sum_{i=1}^{n} b_i^2\Big),$
Minkowski:	$\Big(\sum_{i=1}^{n} (a_i+b_i)^2\Big)^{1/2} \leq \Big(\sum_{i=1}^{n} a_i^2\Big)^{1/2} + \Big(\sum_{i=1}^{n} b_i^2\Big)^{1/2}.$

0.7 Sequences of real numbers

Theorem 0.9.

(i) ***Monotone convergence:*** *a bounded monotone sequence in* $\mathbb{R}$ *converges.*

(ii) ***Bolzano–Weierstrass:*** *every bounded sequence in* $\mathbb{R}$ *(or* $\mathbb{R}^n$*) has a convergent subsequence.*

A **subsequence** of $(x_n)_{n\in\mathbb{N}}$ is $(x_{n_k})_{k\in\mathbb{N}}$ with $n_1 < n_2 < n_3 < \cdots$. In particular $n_k \geq k$ for all k.

Chapter 1

Metric Spaces

Study slot: 2:30–3:05 (35 min)

Learn the three axioms by heart. Work through the standard examples, especially the metrics on $\mathbb{R}^n$, the discrete metric and $C[a,b]$. Practise one "prove this is a metric" proof and one "show this is *not* a metric" counterexample.

Intuition. Analysis on $\mathbb{R}$ (limits, continuity, convergence) only ever uses one thing: the distance $|x-y|$. A *metric space* is any set with a sensible notion of distance. Once we have one, we can do analysis on sets of points, vectors, functions or strings.

1.1 Definition and first examples

Definition 1.1 (Metric space)**.** A **metric** on a set X is a function $d\colon X \times X \to \mathbb{R}$ such that for all $x, y, z \in X$:

(M1) $d(x,y) \geq 0$, and $d(x,y) = 0 \iff x = y$; (positivity)

(M2) $d(x,y) = d(y,x)$; (symmetry)

(M3) $d(x,z) \leq d(x,y) + d(y,z)$. (triangle inequality)

The pair (X,d) is called a **metric space**.

Exam tip. When asked to prove that d is a metric, check (M1), (M2), (M3) explicitly and in order. (M1) and (M2) are usually one line each. All the work goes into (M3).

Example 1.2 (The real line)**.** $X = \mathbb{R}$ with $d(x,y) = |x-y|$, the **usual** (standard) metric.

Example 1.3 (Three metrics on $\mathbb{R}^n$)**.** For $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ in $\mathbb{R}^n$ define

$$d_1(x,y) = \sum_{i=1}^{n} |x_i - y_i| \qquad \text{(taxicab / Manhattan)},$$

$$d_2(x,y) = \Big(\sum_{i=1}^{n} (x_i - y_i)^2\Big)^{1/2} \qquad \text{(Euclidean)},$$

$$d_\infty(x,y) = \max_{1 \leq i \leq n} |x_i - y_i| \qquad \text{(sup / max / Chebyshev)}.$$

(M1) and (M2) are immediate. For d_1 and d_∞, (M3) follows from the triangle inequality in $\mathbb{R}$ applied coordinate by coordinate. For d_2 it is Minkowski's inequality (Exercise 1.5).

Proposition 1.4 (Comparison of the metrics on $\mathbb{R}^n$)**.** *For all $x, y \in \mathbb{R}^n$,*

$$d_\infty(x,y) \;\leq\; d_2(x,y) \;\leq\; d_1(x,y) \;\leq\; n\, d_\infty(x,y), \qquad d_2(x,y) \leq \sqrt{n}\, d_\infty(x,y).$$

Proof. Write $a_i = |x_i - y_i|$ and $M = \max_i a_i$. Then $M^2 \leq \sum a_i^2$, which gives the first inequality.

Expanding $(\sum a_i)^2 = \sum a_i^2 + 2\sum_{i<j} a_i a_j \geq \sum a_i^2$ gives the second. The rest follows from $a_i \leq M$ for every i. □

Example 1.5 (The discrete metric)**.** On *any* set X define

$$d(x,y) = \begin{cases} 0, & x = y, \\ 1, & x \neq y. \end{cases}$$

(M1), (M2) are clear. For (M3): if $x = z$, the left side is 0. If $x \neq z$, then y cannot equal both x and z, so the right side is at least $1 = d(x,z)$.

Warning. The discrete metric is the standard *counterexample machine*. In it, every set is open and closed, every function out of it is continuous, convergent sequences are eventually constant, and the space is complete. It is compact only if X is finite. Whenever you suspect a general statement is false, test it here first.

Example 1.6 (Function spaces)**.** Let $C[a,b]$ be the set of continuous $f\colon [a,b] \to \mathbb{R}$. Define

$$d_\infty(f,g) = \sup_{t\in[a,b]} |f(t) - g(t)|, \qquad d_1(f,g) = \int_a^b |f(t) - g(t)|\, dt.$$

The supremum is finite because $|f-g|$ is continuous on the compact interval $[a,b]$, so it attains a maximum. For d_1, continuity is essential for (M1): if $f \neq g$, then $|f-g| > 0$ on some subinterval, so the integral is positive. Geometrically, $d_\infty(f,g)$ is the largest vertical gap between the graphs, and $d_1(f,g)$ is the area between them.

Example 1.7 (Sequence spaces)**.**

$$\ell^1 = \Big\{x = (x_i) : \sum_i |x_i| < \infty\Big\}, \qquad d(x,y) = \sum_{i=1}^{\infty} |x_i - y_i|,$$

$$\ell^2 = \Big\{x = (x_i) : \sum_i x_i^2 < \infty\Big\}, \qquad d(x,y) = \Big(\sum_{i=1}^{\infty} (x_i - y_i)^2\Big)^{1/2},$$

$$\ell^\infty = \Big\{x = (x_i) : \sup_i |x_i| < \infty\Big\}, \qquad d(x,y) = \sup_i |x_i - y_i|.$$

1.2 Normed spaces

Definition 1.8 (Norm)**.** A **norm** on a real vector space V is a map $\|\cdot\| : V \to \mathbb{R}$ such that for all $u, v \in V$ and $\lambda \in \mathbb{R}$:

(N1) $\|v\| \geq 0$, and $\|v\| = 0 \iff v = 0$;

(N2) $\|\lambda v\| = |\lambda|\, \|v\|$;

(N3) $\|u + v\| \leq \|u\| + \|v\|$.

Every norm induces a metric $d(u,v) = \|u - v\|$.

Remark 1.9. All the metrics in Examples 1.3 and 1.6 come from norms. The discrete metric does *not* come from a norm on any non-zero vector space: (N2) would force $d(\lambda v, 0) = |\lambda|\, d(v,0)$ to take every value in $[0,\infty)$.

1.3 Further examples and non-examples

Example 1.10.

(a) **Hamming metric.** On strings of length n, $d(x,y) = \#\{\, i \,:\, x_i \neq y_i \,\}$.

(b) **Bounded metric.** If d is a metric, then so is $\rho = \dfrac{d}{1+d}$, and $\rho < 1$ (Proposition 1.18).

(c) **British Rail metric** on $\mathbb{R}^2$: $d(x,y) = \|x-y\|$ if x, y and 0 are collinear, and $d(x,y) = \|x\| + \|y\|$ otherwise.

(d) **Subspace metric.** If $Y \subseteq X$, the restriction $d|_{Y\times Y}$ is a metric on Y. For example, $\mathbb{Q}$ or $[0,1]$ with $|x-y|$.

(e) **Product metric.** On $X \times Y$: $d((x,y),(x',y')) = \max\{d_X(x,x'), d_Y(y,y')\}$.

Example 1.11 (Things that are *not* metrics)**.**

(a) $d(x,y) = (x-y)^2$ on $\mathbb{R}$ fails (M3): $d(0,2) = 4 > 2 = d(0,1) + d(1,2)$.

(b) $d(x,y) = |x^2 - y^2|$ on $\mathbb{R}$ fails (M1): $d(1,-1) = 0$ but $1 \neq -1$.

(c) $d(f,g) = |f(0) - g(0)|$ on $C[0,1]$ fails (M1). This is a **pseudometric**.

To disprove an axiom, one explicit numerical counterexample is enough.

1.4 Consequences of the axioms

Lemma 1.12. *For all $x, y, z, x', y' \in X$:*

(i) $|d(x,z) - d(y,z)| \leq d(x,y)$; *(reverse triangle inequality)*

(ii) $d(x_1, x_n) \leq \sum_{k=1}^{n-1} d(x_k, x_{k+1})$; *(polygon inequality)*

(iii) $|d(x,y) - d(x',y')| \leq d(x,x') + d(y,y')$.

Proof. (i) By (M3), $d(x,z) \leq d(x,y) + d(y,z)$, so $d(x,z) - d(y,z) \leq d(x,y)$. Swapping x and y and using (M2) gives $d(y,z) - d(x,z) \leq d(x,y)$. (ii) Induction on n. (iii) Apply (i) twice. □

Definition 1.13 (Bounded sets, diameter, distance to a set)**.** Let $A, B \subseteq X$ be non-empty.

(i) A is **bounded** if there are $x_0 \in X$ and $M \geq 0$ with $d(a, x_0) \leq M$ for all $a \in A$.

(ii) The **diameter** of A is $\operatorname{diam} A = \sup\{\, d(a,b) \,:\, a,b \in A \,\}$. So A is bounded $\iff$ $\operatorname{diam} A < \infty$.

(iii) The **distance** from x to A is $d(x,A) = \inf\{\, d(x,a) \,:\, a \in A \,\}$, and $d(A,B) = \inf\{\, d(a,b) \,:\, a \in A,\ b \in B \,\}$.

Warning. $d(x,A) = 0$ does *not* imply $x \in A$. For example, $d(0,(0,1)) = 0$. It means $x \in \operatorname{cl} A$ (Theorem 2.9). Boundedness also depends on the metric: $\mathbb{R}$ is unbounded for $|x-y|$ but bounded for the discrete metric.

Definition 1.14 (Isometry)**.** A map $f\colon (X, d_X) \to (Y, d_Y)$ is an **isometry** if $d_Y(f(x), f(y)) = d_X(x,y)$ for all $x, y \in X$. Isometries are injective by (M1).

1.5 Equivalent metrics

Definition 1.15. Two metrics d, d' on X are

(i) **Lipschitz** (strongly) **equivalent** if there are constants $\alpha, \beta > 0$ with

$$\alpha\, d(x,y) \le d'(x,y) \le \beta\, d(x,y) \qquad \text{for all } x, y \in X;$$

(ii) **topologically equivalent** if they have the same open sets (Chapter 2).

Proposition 1.16. *Lipschitz equivalent $\Rightarrow$ topologically equivalent, but not conversely.*

Counterexample to the converse. On $\mathbb{R}$, $d = |x-y|$ and $\rho = d/(1+d)$ have the same open sets, because their small balls coincide up to rescaling of the radius. But $d \le \beta\rho < \beta$ would make d bounded, which is false. □

Example 1.17 (d_1 and d_∞ on $C[0,1]$ are not equivalent)**.** Always $d_1(f,g) \le d_\infty(f,g)$, because the area is at most height $\times$ width 1. But for $f_n(t) = t^n$,

$$d_1(f_n, 0) = \int_0^1 t^n\, dt = \frac{1}{n+1} \longrightarrow 0, \qquad d_\infty(f_n, 0) = 1 \ \text{ for all } n.$$

So no inequality $d_\infty \le \beta\, d_1$ can hold.

1.6 Worked proof: $d/(1+d)$ is a metric

Proposition 1.18. *If d is a metric on X, then $\rho(x,y) = \dfrac{d(x,y)}{1+d(x,y)}$ is a metric on X.*

Proof. Let $\varphi(t) = \dfrac{t}{1+t} = 1 - \dfrac{1}{1+t}$ for $t \ge 0$. Then $\varphi \ge 0$, $\varphi(t) = 0 \iff t = 0$, and φ is increasing. So (M1) and (M2) for ρ follow from those for d. Next, φ is *subadditive*: for $a, b \ge 0$,

$$\varphi(a+b) = \frac{a}{1+a+b} + \frac{b}{1+a+b} \le \frac{a}{1+a} + \frac{b}{1+b} = \varphi(a) + \varphi(b).$$

Hence, using (M3) for d and that φ is increasing,

$$\rho(x,z) = \varphi\big(d(x,z)\big) \le \varphi\big(d(x,y) + d(y,z)\big) \le \varphi\big(d(x,y)\big) + \varphi\big(d(y,z)\big) = \rho(x,y) + \rho(y,z). \qquad \square$$

Remark 1.19 (General recipe). If $\varphi\colon [0,\infty) \to [0,\infty)$ is increasing, subadditive and satisfies $\varphi(t) = 0 \iff t = 0$, then $\varphi \circ d$ is a metric. Examples: $\sqrt{d}$, $\min\{d, 1\}$, $d/(1+d)$. This does *not* work for d^2, which is not subadditive.

1.7 Exercises

Exercise 1.1. [Easy] Show that $d(x,y) = |x^3 - y^3|$ is a metric on $\mathbb{R}$.

Solution. (M1): $|x^3 - y^3| \ge 0$, and it equals $0 \iff x^3 = y^3 \iff x = y$, since $t \mapsto t^3$ is injective. (M2) is clear. (M3): $|x^3 - z^3| = |(x^3 - y^3) + (y^3 - z^3)| \le |x^3 - y^3| + |y^3 - z^3|$. In general, $|f(x) - f(y)|$ is a metric whenever f is injective. □

Exercise 1.2. [Easy] Compute d_1, d_2, d_∞ between $(1,2)$ and $(4,-2)$ in $\mathbb{R}^2$.

Solution. The coordinate differences are 3 and 4, so $d_1 = 7$, $d_2 = 5$, $d_\infty = 4$. Check: $4 \le 5 \le 7 \le 2 \cdot 4$. $\square$

Exercise 1.3. [Medium] Is $d(x,y) = \min\{1, |x-y|\}$ a metric on $\mathbb{R}$? Is it Lipschitz equivalent to the usual metric?

Solution. $\varphi(t) = \min\{1,t\}$ is increasing, $\varphi(t) = 0 \iff t = 0$, and it is subadditive. If $a \ge 1$ or $b \ge 1$, the right side is $\ge 1 \ge \varphi(a+b)$; otherwise $\varphi(a+b) \le a+b = \varphi(a)+\varphi(b)$. So d is a metric. It is *not* Lipschitz equivalent to $|x-y|$: we would need $|x-y| \le \beta \min\{1,|x-y|\} \le \beta$ for all x, y. It is topologically equivalent, since balls of radius < 1 coincide. $\square$

Exercise 1.4. [Hard] Fix a prime p. For $0 \neq m \in \mathbb{Z}$, let $v(m)$ be the largest k with $p^k \mid m$. Define $d(x,y) = p^{-v(x-y)}$ for $x \neq y$ and $d(x,x) = 0$. Show that

$$d(x,z) \le \max\{d(x,y),\, d(y,z)\},$$

the **ultrametric inequality**, and deduce that d is a metric on $\mathbb{Z}$.

Solution. If p^k divides both $x-y$ and $y-z$, it divides their sum $x-z$. Hence $v(x-z) \ge \min\{v(x-y), v(y-z)\}$, and so $d(x,z) \le \max\{d(x,y), d(y,z)\}$. (The cases where two of the points coincide are trivial.) Since $\max\{a,b\} \le a+b$, (M3) follows. (M1) and (M2) are clear because $v(-m) = v(m)$. A curious consequence is that in an ultrametric space, every point of a ball is a centre of that ball. $\square$

Exercise 1.5. [Hard] Prove that d_2 satisfies the triangle inequality on $\mathbb{R}^n$.

Solution. Put $a = x-y$ and $b = y-z$, so that $x - z = a + b$. Writing $\|a\| = (\sum a_i^2)^{1/2}$ and using Cauchy–Schwarz, $\sum a_i b_i \le \|a\|\,\|b\|$:

$$\|a+b\|^2 = \sum a_i^2 + 2\sum a_i b_i + \sum b_i^2 \le \|a\|^2 + 2\,\|a\|\,\|b\| + \|b\|^2 = \big(\|a\| + \|b\|\big)^2.$$

Both sides are non-negative, so take square roots. $\square$

Chapter 2

Open and Closed Sets, Convergence, Topological Spaces

Study slot: 3:15–3:50 (35 min)

This is the chapter with the most terminology, and every later chapter uses it. Learn the definitions of open, closed, interior, closure, boundary, limit point and dense word for word. Fill in Example 2.12 yourself before reading the answers.

2.1 Balls

Definition 2.1 (Balls and spheres)**.** For $x \in X$ and $r > 0$:

$$B(x,r) = \{\, y \in X \;:\; d(x,y) < r \,\},$$
$$\overline{B}(x,r) = \{\, y \in X \;:\; d(x,y) \le r \,\},$$
$$S(x,r) = \{\, y \in X \;:\; d(x,y) = r \,\}.$$

These are the **open ball**, the **closed ball** and the **sphere**.

Balls depend on the metric. Figure 2.1 shows the unit ball $B(0,1)$ in $\mathbb{R}^2$ for the three metrics of Example 1.3.

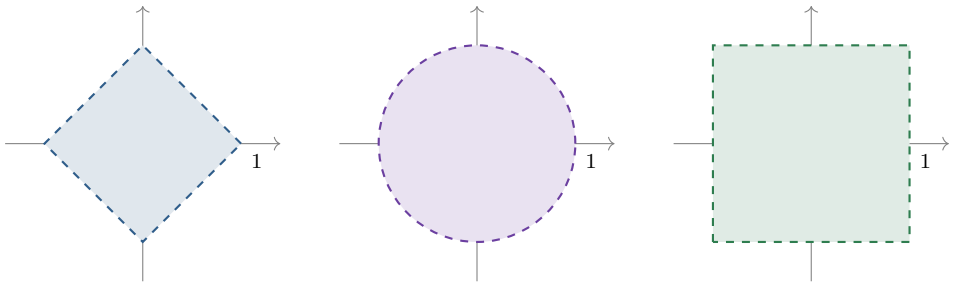


Figure 2.1: Unit open balls in $\mathbb{R}^2$. The dashed boundary is not part of the open ball. The nesting diamond $\subseteq$ disc $\subseteq$ square is the inequality $d_\infty \le d_2 \le d_1$ of Proposition 1.4.

Example 2.2 (Balls in a discrete space)**.** For the discrete metric, $B(x,r) = \{x\}$ if $r \le 1$ and $B(x,r) = X$ if $r > 1$. In particular $B(x,1) = \{x\}$ but $\overline{B}(x,1) = X$.

2.2 Open and closed sets

Definition 2.3 (Open set)**.** A subset $U \subseteq X$ is **open** if

$$\forall x \in U \;\; \exists r > 0 : \quad B(x,r) \subseteq U.$$

Intuition. Every point of U has some breathing room. You can move a little in any direction and remain in U. No point of U sits on its edge.

Definition 2.4 (Closed set)**.** A subset $F \subseteq X$ is **closed** if its complement $X \setminus F$ is open.

Lemma 2.5 (Open balls are open)**.** *Every open ball $B(x,r)$ is open, and every closed ball $\overline{B}(x,r)$ and sphere $S(x,r)$ is closed.*

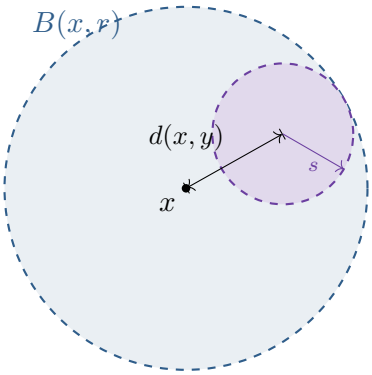


Figure 2.2: The proof of Lemma 2.5: a ball of radius $s = r - d(x,y)$ around y fits inside $B(x,r)$.

Proof. Let $y \in B(x,r)$ and set $s = r - d(x,y) > 0$. If $z \in B(y,s)$, then

$$d(x,z) \le d(x,y) + d(y,z) < d(x,y) + s = r,$$

so $B(y,s) \subseteq B(x,r)$. For the closed ball, we show its complement is open. If $d(x,y) > r$, set $s = d(x,y) - r > 0$. For $z \in B(y,s)$, Lemma 1.12 gives $d(x,z) \ge d(x,y) - d(y,z) > d(x,y) - s = r$. The sphere is $\overline{B}(x,r) \setminus B(x,r) = \overline{B}(x,r) \cap (X \setminus B(x,r))$, an intersection of two closed sets. □

Exam tip. The radius choice $s = r - d(x,y)$ is the most reused idea in the course. Draw the picture.

Theorem 2.6 (Properties of open sets)**.** *In any metric space (X,d):*

(i) $\emptyset$ and X are open;

(ii) an arbitrary union $\bigcup_{i \in I} U_i$ of open sets is open;

(iii) a finite *intersection $U_1 \cap \cdots \cap U_n$ of open sets is open.*

Dually (by De Morgan): $\emptyset$ and X are closed, arbitrary intersections of closed sets are closed, and finite unions of closed sets are closed.

Proof. (i) is vacuous for $\emptyset$, and holds for X with any r. (ii) If $x \in \bigcup U_i$, then $x \in U_j$ for some j, so $B(x,r) \subseteq U_j \subseteq \bigcup U_i$ for some $r > 0$. (iii) If $x \in \bigcap_{k=1}^{n} U_k$, pick $r_k > 0$ with $B(x,r_k) \subseteq U_k$ and let $r = \min\{r_1, \dots, r_n\}$. Then $r > 0$, *because there are finitely many r_k*, and $B(x,r) \subseteq \bigcap U_k$. □

Warning. Finiteness in (iii) is essential:

$$\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\} \ \text{ is not open,} \qquad \bigcup_{n=1}^{\infty} \left[\frac{1}{n}, 1\right] = (0,1] \ \text{ is not closed.}$$

Warning (Sets are not doors). "Not open" does *not* mean "closed". A set can be both open and closed (**clopen**): $\emptyset$ and X always, and every subset of a discrete space. A set can also

be neither: $[0,1)$ or $\mathbb{Q}$ in $\mathbb{R}$.

Example 2.7 (Subsets of $\mathbb{R}$ with the usual metric)**.**

Set	Open	Closed	Reason
(a,b)	✓	✗	It is a ball. A sequence $x_n \to a$ inside it has its limit outside.
$[a,b]$	✗	✓	Every ball around a leaves the set. Its complement is a union of open intervals.
$[a,b)$	✗	✗	Openness fails at a and closedness fails at b.
$\{p\}$, finite sets	✗	✓	Finite sets are closed in every metric space.
$\mathbb{Z}$	✗	✓	Its complement is $\bigcup_n (n, n+1)$.
$\mathbb{Q}$	✗	✗	Every interval contains both rationals and irrationals.
$\{\,1/n \,:\, n \in \mathbb{N}\,\}$	✗	✗	$1/n \to 0 \notin$ set. Adding 0 makes it closed.
$\mathbb{R}$, $\emptyset$	✓	✓	These are always clopen.

Warning (Open *in what*?). $[0,1)$ is not open in $\mathbb{R}$, but it *is* open in the subspace $X = [0,2]$, since $[0,1) = B_X(0,1)$. See Section 2.6.

2.3 Interior, closure, boundary and limit points

Definition 2.8. Let $A \subseteq X$.

(i) x is an **interior point** of A if $B(x,r) \subseteq A$ for some $r > 0$. The **interior** $\operatorname{int} A$ (also written A°) is the set of interior points. Equivalently,

$$\operatorname{int} A = \bigcup \{\, U \,:\, U \subseteq A,\ U \text{ open} \,\},$$

the largest open subset of A.

(ii) The **closure** of A is

$$\operatorname{cl} A = \overline{A} = \bigcap \{\, F \,:\, F \supseteq A,\ F \text{ closed} \,\},$$

the smallest closed superset of A.

(iii) x is a **limit point** (accumulation point) of A if every ball $B(x,r)$ contains a point of $A \setminus \{x\}$. The set A' of limit points is the **derived set**.

(iv) $a \in A$ is an **isolated point** of A if $B(a,r) \cap A = \{a\}$ for some $r > 0$.

(v) The **boundary** is $\partial A = \operatorname{cl} A \setminus \operatorname{int} A$, and the **exterior** is $\operatorname{ext} A = \operatorname{int}(X \setminus A)$.

Theorem 2.9 (Characterisations of the closure)**.** *For $A \subseteq X$ and $x \in X$, the following are equivalent:*

(1) $x \in \operatorname{cl} A$*;*

(2) $B(x,r) \cap A \neq \emptyset$ *for every* $r > 0$*;*

(3) there is a sequence (a_n) in A with $a_n \to x$;

(4) $d(x, A) = 0$;

(5) $x \in A \cup A'$.

Moreover, A is open $\iff$ $A = \operatorname{int} A$, and A is closed $\iff$ $A = \operatorname{cl} A \iff A' \subseteq A$.

Proof. (1)⇒(2): if $B(x,r) \cap A = \emptyset$, then $X \setminus B(x,r)$ is a closed set containing A, hence containing $\operatorname{cl} A$, so $x \notin \operatorname{cl} A$. (2)⇒(3): choose $a_n \in B(x, 1/n) \cap A$. Then $d(a_n, x) < 1/n \to 0$. (3)⇒(4): $0 \le d(x, A) \le d(x, a_n) \to 0$. (4)⇒(2): by Proposition 0.8, for each $r > 0$ some $a \in A$ has $d(x, a) < r$. (2)⇒(1): let E be the set of x satisfying (2). If $y \notin E$, then some $B(y, r)$ misses A. Every $z \in B(y, r)$ has a ball inside $B(y, r)$ (Lemma 2.5), so $z \notin E$. Hence $X \setminus E$ is open, and E is a closed set containing A. Therefore $\operatorname{cl} A \subseteq E$. (2)⇔(5): immediate from the definitions. □

Proposition 2.10 (Useful identities)**.** *For $A, B \subseteq X$:*

$$X \setminus \operatorname{cl} A = \operatorname{int}(X \setminus A), \qquad X \setminus \operatorname{int} A = \operatorname{cl}(X \setminus A),$$
$$\partial A = \operatorname{cl} A \cap \operatorname{cl}(X \setminus A) = \partial(X \setminus A), \qquad \partial A \text{ is closed},$$
$$\operatorname{cl}(A \cup B) = \operatorname{cl} A \cup \operatorname{cl} B, \qquad \operatorname{cl}(A \cap B) \subseteq \operatorname{cl} A \cap \operatorname{cl} B,$$
$$\operatorname{int}(A \cap B) = \operatorname{int} A \cap \operatorname{int} B, \qquad \operatorname{int}(A \cup B) \supseteq \operatorname{int} A \cup \operatorname{int} B.$$

The inclusions can be strict. For $A = \mathbb{Q}$ and $B = \mathbb{R} \setminus \mathbb{Q}$, $\operatorname{cl}(A \cap B) = \emptyset$ but $\operatorname{cl} A \cap \operatorname{cl} B = \mathbb{R}$.

Warning ($\operatorname{cl} B(x,r) \neq \overline{B}(x,r)$ in general). Always $\operatorname{cl} B(x,r) \subseteq \overline{B}(x,r)$. In a discrete space with at least two points, $\operatorname{cl} B(x,1) = \{x\} \neq X = \overline{B}(x,1)$. Equality does hold in every normed space.

Definition 2.11 (Dense, separable, nowhere dense)**.**

(i) A is **dense** in X if $\operatorname{cl} A = X$. Equivalently, every non-empty open set meets A.

(ii) X is **separable** if it has a countable dense subset.

(iii) A is **nowhere dense** if $\operatorname{int}(\operatorname{cl} A) = \emptyset$.

For example, $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$, and $\mathbb{R}^n$ is separable because $\mathbb{Q}^n$ is dense. Polynomials are dense in $(C[a,b], d_\infty)$ (Weierstrass). An uncountable discrete space is not separable. $\mathbb{Z}$ and the Cantor set are nowhere dense in $\mathbb{R}$.

Example 2.12 (Computing everything)**.** Let $A = [0,1) \cup \{2\} \cup (\mathbb{Q} \cap (3,4)) \subseteq \mathbb{R}$.

Object	Answer	Reason
$\operatorname{int} A$	$(0,1)$	Points of $(0,1)$ have room around them. Balls around 0 or 2 leave A. Every interval around a rational contains irrationals.
$\operatorname{cl} A$	$[0,1] \cup \{2\} \cup [3,4]$	1 is a limit of points in A, and $\mathbb{Q}$ is dense in $[3,4]$.
∂A	$\{0,1,2\} \cup [3,4]$	$\operatorname{cl} A \setminus \operatorname{int} A$.
A'	$[0,1] \cup [3,4]$	2 is isolated, so it is not a limit point.
Isolated points	$\{2\}$	$B(2, \frac{1}{2}) \cap A = \{2\}$.
Open / closed?	neither	$0 \in A$ is not interior, and $1 \in \operatorname{cl} A \setminus A$.

2.4 Convergence of sequences

Definition 2.13 (Convergence)**.** A sequence $(x_n)_{n\in\mathbb{N}}$ in X **converges** to $x \in X$, written $x_n \to x$, if $d(x_n, x) \to 0$ in $\mathbb{R}$:

$$\forall \varepsilon > 0 \;\; \exists N \in \mathbb{N} \;\; \forall n \geq N : \quad d(x_n, x) < \varepsilon.$$

Equivalently, every open set containing x contains all but finitely many terms x_n.

Theorem 2.14 (Basic facts)**.**

(i) Limits in a metric space are unique.

(ii) A convergent sequence is bounded.

(iii) If $x_n \to x$, then every subsequence converges to x.

(iv) ***Sequential characterisation of closed sets:*** *F is closed $\iff$ whenever $x_n \in F$ and $x_n \to x \in X$, we have $x \in F$.*

Proof. (i) If $x_n \to x$ and $x_n \to y$, then $0 \leq d(x, y) \leq d(x, x_n) + d(x_n, y) \to 0$, so $x = y$. (ii) With $\varepsilon = 1$, all but finitely many terms lie in $B(x, 1)$. (iii) $n_k \geq k$. (iv) ($\Rightarrow$) If $x \notin F$, some $B(x, r) \subseteq X \setminus F$, so $d(x_n, x) \geq r$ for all n, which is a contradiction. ($\Leftarrow$) By Theorem 2.9(3), each $x \in \operatorname{cl} F$ is a limit of points of F, so $x \in F$. Thus $\operatorname{cl} F = F$. □

Example 2.15 (What convergence means in specific spaces)**.**

- In $(\mathbb{R}^n, d_p)$ for $p = 1, 2, \infty$: convergence $\iff$ coordinatewise convergence.
- In $(C[a, b], d_\infty)$: convergence $\iff$ **uniform convergence**.
- In $(C[a, b], d_1)$: convergence in mean, which is strictly weaker (Example 1.17).
- In a discrete space: $x_n \to x \iff x_n = x$ for all sufficiently large n (take $\varepsilon = \frac{1}{2}$).

2.5 Topological spaces

Intuition. Convergence, closure and (as we will see) continuity can all be described using only *which sets are open*. So we can forget the distance and keep just the collection of open sets. That collection is a topology.

Definition 2.16 (Topology)**.** A **topology** on a set X is a collection τ of subsets of X such that

(T1) $\emptyset \in \tau$ and $X \in \tau$;

(T2) if $U_i \in \tau$ for all $i \in I$, then $\bigcup_{i\in I} U_i \in \tau$;

(T3) if $U, V \in \tau$, then $U \cap V \in \tau$.

The members of τ are the **open sets**, and (X, τ) is a **topological space**. F is **closed** if $X \setminus F \in \tau$. A **neighbourhood** of x is a set N with $x \in U \subseteq N$ for some $U \in \tau$.

Example 2.17 (Standard topologies)**.**

(a) The **metric topology** τ_d: the open sets of (X, d). Theorem 2.6 is exactly (T1)–(T3).

(b) The **discrete topology** $\tau = \mathcal{P}(X)$, which comes from the discrete metric.

(c) The **indiscrete** (trivial) topology $\tau = \{\emptyset, X\}$.

(d) The **cofinite topology** $\tau = \{\emptyset\} \cup \{\, U \subseteq X \;:\; X \setminus U \text{ is finite} \,\}$.

(e) **Sierpiński space**: $X = \{0,1\}$, $\tau = \{\emptyset, \{1\}, \{0,1\}\}$.

Example 2.18. $\tau = \{\emptyset, \{a\}, \{b\}, \{a,b,c\}\}$ is *not* a topology on $\{a,b,c\}$, since $\{a\} \cup \{b\} = \{a,b\} \notin \tau$.

Definition 2.19.

(i) τ_1 is **coarser** than τ_2 (and τ_2 is **finer**) if $\tau_1 \subseteq \tau_2$.

(ii) A **basis** for τ is a collection $\mathcal{B} \subseteq \tau$ such that every open set is a union of members of $\mathcal{B}$. For example, open balls form a basis for τ_d.

(iii) (X, τ) is **metrizable** if $\tau = \tau_d$ for some metric d.

(iv) (X, τ) is **Hausdorff** if for all $x \neq y$ there are disjoint open $U \ni x$ and $V \ni y$.

Theorem 2.20. *Every metric space is Hausdorff.*

Proof. If $x \neq y$, let $r = \frac{1}{2}d(x,y) > 0$. If $z \in B(x,r) \cap B(y,r)$, then $d(x,y) \leq d(x,z) + d(z,y) < 2r = d(x,y)$, a contradiction. □

Exam tip. To show a topology is *not metrizable*, show it is not Hausdorff. This covers the indiscrete topology on ≥ 2 points, the cofinite topology on an infinite set, and Sierpiński space.

Warning. In non-Hausdorff spaces, limits need not be unique. In the indiscrete topology, every sequence converges to every point. Uniqueness of limits really comes from the Hausdorff property.

2.6 Subspaces

Definition 2.21 (Subspace topology)**.** If $Y \subseteq X$, the **subspace topology** is $\tau_Y = \{\, U \cap Y \;:\; U \in \tau \,\}$. For a metric space this agrees with the topology of the restricted metric, and $B_Y(y,r) = B_X(y,r) \cap Y$.

Proposition 2.22. *$F \subseteq Y$ is closed in $Y \iff F = C \cap Y$ for some C closed in X. If Y is open (respectively closed) in X, a subset of Y is open (respectively closed) in $Y \iff$ it is open (respectively closed) in X.*

Example 2.23. In $Y = \mathbb{Q}$, the set $\{\, q \in \mathbb{Q} \;:\; q^2 < 2 \,\} = (-\sqrt{2}, \sqrt{2}) \cap \mathbb{Q} = [-\sqrt{2}, \sqrt{2}] \cap \mathbb{Q}$ is clopen.

2.7 Exercises

Exercise 2.1. [Easy] Prove directly that $U = \{\,(x,y) \in \mathbb{R}^2 \;:\; x > 0\,\}$ is open.

Solution. Let $p = (a,b) \in U$ and take $r = a > 0$. If $(x,y) \in B(p,a)$, then $|x - a| \le d_2((x,y),p) < a$, so $x > 0$. Hence $B(p,a) \subseteq U$. (Faster, using Chapter 3: $U = f^{-1}((0,\infty))$ for the continuous map $f(x,y) = x$.) □

Exercise 2.2. [Easy] Prove that every finite subset of a metric space is closed.

Solution. If $x \neq p$, then $B\big(x, d(x,p)\big)$ does not contain p, so $X \setminus \{p\}$ is open and $\{p\}$ is closed. A finite set is a finite union of singletons, so it is closed by Theorem 2.6. □

Exercise 2.3. [Medium] In a discrete metric space, find $\operatorname{int} A$, $\operatorname{cl} A$ and ∂A for an arbitrary A.

Solution. $B(x, \tfrac{1}{2}) = \{x\} \subseteq A$ for every $x \in A$, so every set is open. Hence every set is also closed. Therefore $\operatorname{int} A = A = \operatorname{cl} A$ and $\partial A = \emptyset$. □

Exercise 2.4. [Medium] Show that $\partial A = \emptyset \iff A$ is clopen.

Solution. $\partial A = \operatorname{cl} A \setminus \operatorname{int} A = \emptyset \iff \operatorname{cl} A \subseteq \operatorname{int} A$. Since $\operatorname{int} A \subseteq A \subseteq \operatorname{cl} A$ always, this holds $\iff \operatorname{int} A = A = \operatorname{cl} A$. □

Exercise 2.5. [Hard] Let X be infinite with the cofinite topology. (a) Verify (T1)–(T3). (b) Show X is not Hausdorff. (c) Determine which sequences converge, and to what.

Solution. (a) $X \setminus X = \emptyset$ is finite. For non-empty open U_i, $X \setminus \bigcup U_i = \bigcap (X \setminus U_i)$ is finite. Also $X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$ is finite. (b) If U, V are non-empty, open and disjoint, then $X = (X \setminus U) \cup (X \setminus V)$ is finite, a contradiction. (c) If no value occurs infinitely often (for example, if all x_n are distinct), then $x_n \to x$ for *every* x: an open $U \ni x$ misses only finitely many points, hence only finitely many terms. If exactly one value a occurs infinitely often, then $x_n \to a$ and to no other point. For $c \neq a$, the open set $X \setminus \{a\} \ni c$ misses infinitely many terms. If two or more values occur infinitely often, the sequence converges to nothing. □

Exercise 2.6. [Hard] Prove $\operatorname{cl} B(x,r) \subseteq \overline{B}(x,r)$ and give two examples where the inclusion is strict.

Solution. $\overline{B}(x,r)$ is closed (Lemma 2.5) and contains $B(x,r)$, and the closure is the smallest such set. Strict examples: (a) the discrete metric with $r = 1$; (b) $X = [0,1] \cup [2,3] \subseteq \mathbb{R}$, $x = 0$, $r = 2$, where $\operatorname{cl} B(0,2) = [0,1]$ but $\overline{B}(0,2) = [0,1] \cup \{2\}$. □

Chapter 3

Continuity and Homeomorphisms

Study slot: 3:50–4:25 (35 min)

Learn the four equivalent definitions of continuity (Theorem 3.2) and when to use each. Memorise the hierarchy Lipschitz ⇒ uniformly continuous ⇒ continuous, with a counterexample for each converse. Be able to prove that two spaces are, or are not, homeomorphic.

Intuition. f is continuous if close inputs give close outputs. To make $f(y)$ within ε of $f(x)$, it is enough to take y within δ of x.

3.1 Four equivalent definitions

Definition 3.1 (ε–δ continuity)**.** Let $f\colon (X, d_X) \to (Y, d_Y)$. Then f is **continuous at** $a \in X$ if

$$\forall \varepsilon > 0 \;\; \exists \delta > 0 \;\; \forall x \in X : \quad d_X(x, a) < \delta \implies d_Y\big(f(x), f(a)\big) < \varepsilon.$$

Equivalently, $f(B(a,\delta)) \subseteq B(f(a),\varepsilon)$. We call f **continuous** if it is continuous at every point of X.

Theorem 3.2 (Characterisations of continuity)**.** *For $f\colon X \to Y$ between metric spaces, the following are equivalent:*

(1) f is continuous (Definition 3.1) at every point;

*(2) **(sequential)** $x_n \to x$ in X implies $f(x_n) \to f(x)$ in Y;*

*(3) **(open sets)** V open in Y implies $f^{-1}(V)$ open in X;*

*(4) **(closed sets)** C closed in Y implies $f^{-1}(C)$ closed in X.*

For topological spaces, (3) is taken as the definition *of continuity.*

Proof. **(1)⇒(3).** Let $V \subseteq Y$ be open and $a \in f^{-1}(V)$. Choose $\varepsilon > 0$ with $B(f(a),\varepsilon) \subseteq V$, and then δ as in (1). Then

$$f(B(a,\delta)) \subseteq B(f(a),\varepsilon) \subseteq V, \quad \text{so} \quad B(a,\delta) \subseteq f^{-1}(V).$$

(3)⇒(1). $B(f(a),\varepsilon)$ is open, so $f^{-1}(B(f(a),\varepsilon))$ is an open set containing a. It therefore contains some $B(a,\delta)$, which is exactly the statement in (1).

(3)⇔(4). Use $f^{-1}(Y \setminus C) = X \setminus f^{-1}(C)$ (Proposition 0.4).

(1)⇒(2). Given $\varepsilon > 0$, take δ from (1) and N with $d(x_n, x) < \delta$ for $n \geq N$. Then $d\big(f(x_n), f(x)\big) < \varepsilon$ for $n \geq N$.

(2)⇒(1) (contrapositive). Suppose (1) fails at a. Then there is $\varepsilon > 0$ such that for each $\delta = 1/n$ there is x_n with $d(x_n, a) < 1/n$ but $d\big(f(x_n), f(a)\big) \geq \varepsilon$. Thus $x_n \to a$ but $f(x_n) \not\to f(a)$. □

Exam tip (Which form to use).

- To show f *is* continuous: use ε–δ, or build f from known continuous maps (Proposition 3.3), or show f is Lipschitz.
- To show f is *not* continuous: give *one* sequence $x_n \to x$ with $f(x_n) \not\to f(x)$.
- To show a set is open or closed: write it as f^{-1}(open) or f^{-1}(closed) for a continuous f.
- In abstract or topological settings: use preimages of open sets.

3.2 Building continuous functions

Proposition 3.3.

(i) Constant maps and the identity $(X,d) \to (X,d)$ are continuous.

(ii) If $f\colon X \to Y$ and $g\colon Y \to Z$ are continuous, so is $g \circ f$, since $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W))$.

(iii) If $f, g\colon X \to \mathbb{R}$ are continuous, so are $f \pm g$, fg, $|f|$, $\max\{f,g\}$, $\min\{f,g\}$, and f/g where $g \neq 0$.

(iv) $f = (f_1, \dots, f_n)\colon X \to \mathbb{R}^n$ is continuous $\iff$ each f_i is continuous.

(v) Every function from a discrete space is continuous.

(vi) For $\emptyset \neq A \subseteq X$, the map $x \mapsto d(x,A)$ is 1-Lipschitz: $|d(x,A) - d(y,A)| \leq d(x,y)$.

(vii) Restrictions of continuous maps to subspaces are continuous.

Example 3.4 (One-line proofs that sets are open or closed)**.**

- $\{\,(x,y) \in \mathbb{R}^2 \;:\; x^2 + y^2 < 1\,\} = f^{-1}((-\infty, 1))$ with $f(x,y) = x^2 + y^2$, so it is open.
- $\{\,(x,y) \in \mathbb{R}^2 \;:\; xy = 1\,\} = g^{-1}(\{1\})$ with $g(x,y) = xy$, so it is closed.
- $\left\{\, f \in C[0,1] \;:\; f(\tfrac{1}{2}) > 0 \,\right\}$ is open in d_∞, because evaluation $E(f) = f(\frac{1}{2})$ satisfies $|E(f) - E(g)| \leq d_\infty(f,g)$.
- The zero set $\{\,x \;:\; f(x) = 0\,\}$ of a continuous $f\colon X \to \mathbb{R}$ is closed.

Warning (Images are badly behaved). Continuous maps need not send open sets to open sets, or closed sets to closed sets:

$$x \mapsto x^2 :\ (-1,1) \longmapsto [0,1), \qquad \arctan :\ \mathbb{R} \longmapsto \left(-\tfrac{\pi}{2}, \tfrac{\pi}{2}\right).$$

Projecting the closed hyperbola $\{xy = 1\}$ to the x-axis gives $\mathbb{R} \setminus \{0\}$, which is not closed. Only *preimages* behave well, apart from compact sets (Theorem 5.7).

Warning (Continuity depends on the metrics). $\mathrm{id}\colon (\mathbb{R}, \text{discrete}) \to (\mathbb{R}, |\cdot|)$ is continuous, but $\mathrm{id}\colon (\mathbb{R}, |\cdot|) \to (\mathbb{R}, \text{discrete})$ is not: $\mathrm{id}^{-1}(\{0\}) = \{0\}$ is not open. Likewise $\mathrm{id}\colon (C[0,1], d_\infty) \to (C[0,1], d_1)$ is continuous, since $d_1 \leq d_\infty$, but its inverse is not (Example 1.17).

3.3 Uniform continuity and Lipschitz maps

Definition 3.5. Let $f\colon (X, d_X) \to (Y, d_Y)$.

(i) f is **uniformly continuous** if

$$\forall \varepsilon > 0 \;\; \exists \delta > 0 \;\; \forall x, y \in X : \quad d_X(x,y) < \delta \implies d_Y\big(f(x), f(y)\big) < \varepsilon.$$

Here δ depends only on ε, not on the point. Compare the position of "$\forall x$" with Definition 3.1.

(ii) f is **Lipschitz** with constant $L \geq 0$ if $d_Y\big(f(x), f(y)\big) \leq L\, d_X(x,y)$ for all $x, y \in X$. If $L < 1$, f is a **contraction** (Chapter 4).

Proposition 3.6. *Lipschitz $\Rightarrow$ uniformly continuous $\Rightarrow$ continuous. Neither implication reverses.*

Proof. If f is L-Lipschitz with $L > 0$, take $\delta = \varepsilon / L$. The second implication is immediate from the definitions. The counterexamples are in the table below. □

Function	Continuous	Uniformly continuous	Lipschitz
x^2 on $\mathbb{R}$	✓	✗	✗
$1/x$ on $(0,1)$	✓	✗	✗
$\sin(1/x)$ on $(0,1)$	✓	✗	✗
$\sqrt{x}$ on $[0,1]$	✓	✓ (compact domain)	✗ (slope $\to \infty$ at 0)
$\sin x$ on $\mathbb{R}$	✓	✓	✓ ($L = 1$)
x^2 on $[0,1]$	✓	✓	✓ ($L = 2$)

Proposition 3.7 (Sequential criterion)**.** *f is* not *uniformly continuous $\iff$ there exist $\varepsilon > 0$ and sequences $(x_n), (y_n)$ with*

$$d(x_n, y_n) \to 0 \quad \text{but} \quad d\big(f(x_n), f(y_n)\big) \geq \varepsilon \;\; \text{for all } n.$$

Example 3.8 (x^2 is not uniformly continuous on $\mathbb{R}$)**.** Take $x_n = n$ and $y_n = n + \frac{1}{n}$. Then $|x_n - y_n| = \frac{1}{n} \to 0$, but

$$y_n^2 - x_n^2 = 2 + \frac{1}{n^2} \geq 2.$$

Proposition 3.9. *A uniformly continuous map sends Cauchy sequences to Cauchy sequences. A continuous map need not: $1/x$ on $(0,1)$ sends the Cauchy sequence $(1/n)$ to (n).*

3.4 Homeomorphisms

Definition 3.10. A **homeomorphism** is a bijection $f\colon X \to Y$ such that both f and f^{-1} are continuous. If one exists, X and Y are **homeomorphic**, written $X \cong Y$. A **topological property** is one preserved by homeomorphisms.

Intuition. A homeomorphism stretches and bends without tearing or gluing. This is why topologists treat a coffee mug and a doughnut as the same space.

Example 3.11.

(a) $(-1,1) \cong \mathbb{R}$ via $x \mapsto \dfrac{x}{1-|x|}$, with inverse $y \mapsto \dfrac{y}{1+|y|}$. Alternatively, use $x \mapsto \tan(\pi x/2)$.

(b) Any two open intervals are homeomorphic, and so are any two closed intervals (use affine maps $x \mapsto \alpha x + \beta$).

(c) An isometry is a homeomorphism onto its image.

(d) If d, d' are topologically equivalent, then $\mathrm{id}\colon (X,d) \to (X,d')$ is a homeomorphism.

Warning (A continuous bijection need not be a homeomorphism). The map $f\colon [0,2\pi) \to S^1$, $f(t) = (\cos t, \sin t)$, is a continuous bijection, but f^{-1} is not continuous at $(1,0)$. Points just below $(1,0)$ on the circle are sent to numbers near 2π, not near 0. Another example is $\mathrm{id}\colon (\mathbb{R}, \text{discrete}) \to (\mathbb{R}, |\cdot|)$. A continuous bijection *is* a homeomorphism when the domain is compact and the codomain is Hausdorff (Theorem 5.10).

Topological (preserved by $\cong$)	**Not topological** (depends on the metric)
open and closed sets, convergence, continuity, Hausdorff, compactness, connectedness, separability, number of isolated points	boundedness ($\mathbb{R} \cong (-1,1)$), completeness ($\mathbb{R} \cong (0,1)$), being Cauchy, uniform continuity, diameter

Exam tip (Proving $X \not\cong Y$). Find a topological property that one space has and the other lacks. $[0,1] \not\cong (0,1)$ because only $[0,1]$ is compact. $\mathbb{R} \not\cong \mathbb{R}^2$ because removing a point disconnects $\mathbb{R}$ but not $\mathbb{R}^2$. $[0,1) \not\cong (0,1)$ because removing 0 leaves $[0,1)$ connected, while removing any point disconnects $(0,1)$.

3.5 Exercises

Exercise 3.1. [Easy] Prove from the definition that $f(x) = 3x + 2$ is continuous on $\mathbb{R}$.

Solution. Given $\varepsilon > 0$, put $\delta = \varepsilon/3$. If $|x-a| < \delta$, then $|f(x) - f(a)| = 3\,|x-a| < \varepsilon$. In fact f is 3-Lipschitz, so it is uniformly continuous. □

Exercise 3.2. [Easy] Let $f = \mathbf{1}_{[0,\infty)}$, that is, $f(x) = 1$ for $x \geq 0$ and $f(x) = 0$ for $x < 0$. Show f is not continuous in two ways.

Solution. Sequences: $x_n = -\frac{1}{n} \to 0$, but $f(x_n) = 0 \not\to 1 = f(0)$. *Open sets:* $f^{-1}((\tfrac{1}{2}, 2)) = [0,\infty)$ is not open. □

Exercise 3.3. [Medium] For $\emptyset \neq A \subseteq X$, prove $|d(x,A) - d(y,A)| \leq d(x,y)$. Deduce that $\operatorname{cl} A = \{\, x \,:\, d(x,A) = 0 \,\}$ is closed.

Solution. For every $a \in A$, $d(x,A) \leq d(x,a) \leq d(x,y) + d(y,a)$. Taking the infimum over a gives $d(x,A) \leq d(x,y) + d(y,A)$. Swapping x and y gives the reverse inequality. So $x \mapsto d(x,A)$ is continuous, and $\operatorname{cl} A$ is the preimage of the closed set $\{0\}$ (Theorem 2.9). □

Exercise 3.4. [Medium] Show that $I\colon C[0,1] \to \mathbb{R}$, $I(f) = \int_0^1 f$, is continuous for both d_∞ and d_1.

Solution. $|I(f) - I(g)| = \left|\int_0^1 (f-g)\right| \le \int_0^1 |f-g| = d_1(f,g) \le d_\infty(f,g)$, so I is 1-Lipschitz for both metrics. □

Exercise 3.5. [Hard] Is $E(f) = f(0)$ continuous from $(C[0,1], d_1)$ to $\mathbb{R}$?

Solution. No. Let $f_n(t) = \max\{0, 1-nt\}$, a spike of height 1 at 0. Then $d_1(f_n, 0) = \frac{1}{2n} \to 0$, so $f_n \to 0$. But $E(f_n) = 1 \not\to 0 = E(0)$. (With d_∞, E is 1-Lipschitz.) □

Exercise 3.6. [Hard] Let $f, g\colon X \to Y$ be continuous, where Y is a metric space. Show $\{\, x \,:\, f(x) = g(x) \,\}$ is closed. Deduce that if $f = g$ on a dense set, then $f = g$.

Solution. $h(x) = d_Y\big(f(x), g(x)\big)$ is continuous, since $x_n \to x$ implies $f(x_n) \to f(x)$ and $g(x_n) \to g(x)$, and then $h(x_n) \to h(x)$ by Lemma 1.12(iii). The set is $h^{-1}(\{0\})$, which is closed. If it contains a dense D, it contains $\operatorname{cl} D = X$. For example, a continuous function on $\mathbb{R}$ is determined by its values on $\mathbb{Q}$. □

Chapter 4

Completeness

Study slot: 4:40–5:10 (30 min)

Learn the definition of a Cauchy sequence, the list of complete and incomplete spaces, Theorem 4.6 (closed $\iff$ complete), and the full proof of the Contraction Mapping Theorem (Theorem 4.9), a favourite long exam question.

Intuition. A sequence is Cauchy if its terms eventually get as close to *each other* as you like, so they look as if they are converging. Completeness says that whenever this happens, there really is a limit in the space. $\mathbb{Q}$ is incomplete because $1,\ 1.4,\ 1.41,\ 1.414, \ldots$ "wants" to converge to $\sqrt{2} \notin \mathbb{Q}$. The space has holes.

4.1 Cauchy sequences

Definition 4.1 (Cauchy sequence)**.** A sequence $(x_n)_{n\in\mathbb{N}}$ in (X, d) is **Cauchy** if

$$\forall \varepsilon > 0 \ \ \exists N \in \mathbb{N} \ \ \forall m, n \geq N : \quad d(x_m, x_n) < \varepsilon.$$

Unlike the definition of convergence, this does not mention a limit.

Proposition 4.2.

(i) Every convergent sequence is Cauchy.

(ii) Every Cauchy sequence is bounded.

(iii) A Cauchy sequence with a convergent subsequence converges, to the same limit.

Proof. (i) If $x_n \to x$, choose N with $d(x_n, x) < \varepsilon/2$ for $n \geq N$. Then for $m, n \geq N$,

$$d(x_m, x_n) \leq d(x_m, x) + d(x, x_n) < \varepsilon.$$

(ii) With $\varepsilon = 1$, all x_n with $n \geq N$ lie in $B(x_N, 1)$. So every term lies in $B(x_N, R)$ with $R = 1 + \max_{i<N} d(x_i, x_N)$. (iii) Let $x_{n_k} \to x$ and $\varepsilon > 0$. Choose N with $d(x_m, x_n) < \varepsilon/2$ for $m, n \geq N$, then k with $n_k \geq N$ and $d(x_{n_k}, x) < \varepsilon/2$. For $n \geq N$,

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) < \varepsilon. \qquad \square$$

4.2 Complete metric spaces

Definition 4.3 (Complete space)**.** (X, d) is **complete** if every Cauchy sequence in X converges to a point *of* X. A complete normed vector space is a **Banach space**.

Example 4.4 (Complete spaces)**.**

(a) $\mathbb{R}$. A Cauchy sequence is bounded, so it has a convergent subsequence by Bolzano–Weierstrass, so it converges by Proposition 4.2(iii). This uses the completeness axiom.

(b) $(\mathbb{R}^n, d_p)$ for $p = 1, 2, \infty$: each coordinate sequence is Cauchy in $\mathbb{R}$.

(c) Every closed subset of $\mathbb{R}^n$, such as $[0,1]$, $\mathbb{Z}$ or the Cantor set (Theorem 4.6).

(d) Every discrete metric space: with $\varepsilon = \frac{1}{2}$, a Cauchy sequence is eventually constant.

(e) $(C[a,b], d_\infty)$ (Theorem 4.7).

(f) $\ell^1, \ell^2, \ell^\infty$, and every compact metric space (Chapter 5).

Example 4.5 (Incomplete spaces and the Cauchy sequence that shows it)**.**

Space	Cauchy sequence with no limit in the space
$\mathbb{Q}$	decimal truncations of $\sqrt{2}$: $1,\ 1.4,\ 1.41, \cdots \to \sqrt{2} \notin \mathbb{Q}$
$(0,1)$	$x_n = \frac{1}{n} \to 0 \notin (0,1)$
$\mathbb{R} \setminus \{0\}$	$x_n = \frac{1}{n}$
$(C[0,1], d_1)$	continuous ramps tending to a step function (Exercise 4.4)
polynomials on $[0,1]$ with d_∞	Taylor partial sums $\sum_{k \le n} t^k/k! \to e^t$, which is not a polynomial
$\mathbb{R}$ with $d(x,y) = \lvert\arctan x - \arctan y\rvert$	$x_n = n$, since $\arctan n \to \frac{\pi}{2}$

Warning (Completeness is not a topological property). $\mathbb{R} \cong (0,1)$, but only $\mathbb{R}$ is complete. The last row above gives $\mathbb{R}$ with its usual topology but an incomplete metric. Completeness *is* preserved by Lipschitz equivalent metrics, since they have the same Cauchy sequences.

Theorem 4.6 (Closed $\iff$ complete)**.**

(i) If X is complete and $F \subseteq X$ is closed, then F is complete.

(ii) If $Y \subseteq X$ is complete (with the induced metric), then Y is closed in X. This holds in any *metric space X.*

In particular, a subset of a complete metric space is complete if and only if it is closed.

Proof. (i) A Cauchy sequence in F is Cauchy in X, so it converges to some $x \in X$. Since F is closed, $x \in F$ by Theorem 2.14(iv). (ii) Let $y_n \in Y$ with $y_n \to x \in X$. Then (y_n) is convergent, hence Cauchy, so $y_n \to y$ for some $y \in Y$. By uniqueness of limits, $x = y \in Y$. □

Theorem 4.7. *$(C[a,b], d_\infty)$ is complete.*

Proof. Let (f_n) be Cauchy for d_∞.

Step 1 (candidate limit). For each $t \in [a,b]$, $|f_n(t) - f_m(t)| \le d_\infty(f_n, f_m)$, so $(f_n(t))$ is Cauchy in $\mathbb{R}$ and converges. Define $f(t) = \lim_n f_n(t)$.

Step 2 (convergence in the metric). Given $\varepsilon > 0$, choose N with $|f_n(t) - f_m(t)| < \varepsilon$ for all t and all $m, n \ge N$. Letting $m \to \infty$ gives $|f_n(t) - f(t)| \le \varepsilon$ for all t and $n \ge N$, that is, $\sup_t |f_n(t) - f(t)| \to 0$.

Step 3 (the limit lies in the space; the $\varepsilon/3$ argument). Fix n with $\sup_t |f_n(t) - f(t)| < \varepsilon/3$, and

$\delta > 0$ with $|f_n(t) - f_n(s)| < \varepsilon/3$ whenever $|t - s| < \delta$. Then for $|t - s| < \delta$,

$$|f(t) - f(s)| \le |f(t) - f_n(t)| + |f_n(t) - f_n(s)| + |f_n(s) - f(s)| < \varepsilon.$$

So $f \in C[a, b]$ and $d_\infty(f_n, f) \to 0$. □

Exam tip (Template for "prove X is complete"). (1) Take a Cauchy sequence and find a candidate limit, usually pointwise or coordinatewise. (2) Show convergence to the candidate *in the metric*. (3) Show the candidate belongs to X.

4.3 The Contraction Mapping Theorem

Definition 4.8. A map $f\colon X \to X$ is a **contraction** if there is a constant $k \in [0, 1)$ with

$$d(f(x), f(y)) \le k\, d(x, y) \qquad \text{for all } x, y \in X.$$

A **fixed point** of f is a point x^* with $f(x^*) = x^*$.

Theorem 4.9 (Contraction Mapping Theorem / Banach Fixed Point Theorem)**.** *Let (X, d) be a* non-empty complete *metric space and $f\colon X \to X$ a contraction with constant k. Then:*

(i) f has a unique *fixed point x^*;*

(ii) for every $x_0 \in X$, the iterates $x_{n+1} = f(x_n)$ converge to x^*;

(iii) (a priori error bound) $d(x_n, x^*) \le \dfrac{k^n}{1 - k}\, d(x_0, x_1)$.

Proof. Step 1. By induction, $d(x_n, x_{n+1}) = d(f(x_{n-1}), f(x_n)) \le k\, d(x_{n-1}, x_n) \le \cdots \le k^n d(x_0, x_1)$.

Step 2 (Cauchy). For $m > n$, by the polygon inequality and the geometric series,

$$d(x_n, x_m) \le \sum_{j=n}^{m-1} d(x_j, x_{j+1}) \le d(x_0, x_1) \sum_{j=n}^{\infty} k^j = \frac{k^n}{1 - k}\, d(x_0, x_1) \xrightarrow[n\to\infty]{} 0,$$

because $0 \le k < 1$.

Step 3 (convergence). X is complete, so $x_n \to x^*$ for some $x^* \in X$.

Step 4 (fixed point). f is Lipschitz, hence continuous, so

$$f(x^*) = f\big(\lim_n x_n\big) = \lim_n f(x_n) = \lim_n x_{n+1} = x^*.$$

Step 5 (uniqueness). If $f(x) = x$ and $f(y) = y$, then $d(x, y) = d(f(x), f(y)) \le k\, d(x, y)$. So $(1 - k)\, d(x, y) \le 0$, giving $d(x, y) = 0$.

Finally, (iii) follows by letting $m \to \infty$ in Step 2, using continuity of d. □

Warning (Every hypothesis is needed).

- *Completeness:* $f(x) = x/2$ on $(0, 1]$ is a contraction with no fixed point.
- *A uniform $k < 1$:* $f(x) = x + \frac{1}{x}$ on $[1, \infty)$ satisfies $|f(x) - f(y)| = |x - y|\,(1 - \frac{1}{xy}) < |x - y|$ for $x \ne y$, but $f(x) = x$ has no solution.
- *$k = 1$ is not enough:* $f(x) = x + 1$ on $\mathbb{R}$.

- $f(X) \subseteq X$*:* always check this first in applications.

Example 4.10 (Solving $\cos x = x$)**.** Take $X = [0,1]$, which is closed in $\mathbb{R}$ and hence complete, and $f = \cos$. Then:

(i) $f(X) = [\cos 1, 1] \subseteq [0,1]$;

(ii) by the Mean Value Theorem, $|\cos x - \cos y| = |\sin \xi|\,|x-y| \le \sin(1)\,|x-y|$, and $k = \sin 1 \approx 0.84 < 1$.

Therefore $\cos x = x$ has exactly one solution in $[0,1]$ (namely $x^* \approx 0.739$), and iterating cos from any starting point converges to it.

Remark 4.11 (Further applications). The Picard–Lindelöf theorem on existence and uniqueness for $y' = F(t,y)$ applies Theorem 4.9 to the integral operator

$$(Ty)(t) = y_0 + \int_{t_0}^{t} F(s, y(s))\, ds$$

on a closed subset of $(C[a,b], d_\infty)$. Other applications include integral equations, the convergence of Newton's method, and the inverse function theorem.

Exam tip (Recipe for applying the theorem). (1) Name a complete space X and justify it. (2) Check $f(X) \subseteq X$. (3) Find $k < 1$, typically using $\sup |f'| < 1$ and the Mean Value Theorem. (4) Conclude there is a unique fixed point, which is the limit of the iterates.

4.4 Further results

Theorem 4.12 (Cantor Intersection Theorem)**.** *Let X be complete and $F_1 \supseteq F_2 \supseteq \cdots$ non-empty closed subsets with* $\operatorname{diam} F_n \to 0$*. Then $\bigcap_{n=1}^{\infty} F_n$ consists of exactly one point.*

Proof. Pick $x_n \in F_n$. For $m \ge n$, both x_m and x_n lie in F_n, so $d(x_m, x_n) \le \operatorname{diam} F_n \to 0$. Thus (x_n) is Cauchy and converges to some x. Each F_n is closed and contains the tail $(x_m)_{m \ge n}$, so $x \in F_n$ for every n. If y also lies in the intersection, then $d(x,y) \le \operatorname{diam} F_n \to 0$, so $y = x$. □

The hypotheses are needed: $\bigcap_n (0, \frac{1}{n}) = \emptyset$ (not closed), and $\bigcap_n [n, \infty) = \emptyset$ (the diameters do not tend to 0).

Theorem 4.13 (Baire Category Theorem)**.** *In a complete metric space, every countable intersection of dense open sets is dense. Equivalently, a non-empty complete metric space is not a countable union of nowhere dense sets.*

Consequences: $\mathbb{R}$ is uncountable; $\mathbb{Q}$ is not complete under any metric topologically equivalent to the usual one; continuous nowhere-differentiable functions exist.

Theorem 4.14 (Completion)**.** *Every metric space X embeds isometrically as a dense subset of a complete metric space $\widehat{X}$, which is unique up to isometry. For example, $\widehat{\mathbb{Q}} = \mathbb{R}$.*

4.5 Exercises

Exercise 4.1. [Easy] Are $[0,\infty)$, $\mathbb{Z}$ and $\left\{ \frac{1}{n} \,:\, n \in \mathbb{N} \right\}$ complete?

Solution. $[0,\infty)$: yes, it is closed in $\mathbb{R}$. $\mathbb{Z}$: yes, it is closed, and its Cauchy sequences are eventually constant. $\left\{\frac{1}{n} : n \in \mathbb{N}\right\}$: no, because $(\frac{1}{n})$ is Cauchy with limit 0 outside the set. □

Exercise 4.2. [Easy] Show that $x_n = \sum_{k=1}^{n} \frac{1}{k}$ is not Cauchy.

Solution. $x_{2n} - x_n = \displaystyle\sum_{k=n+1}^{2n} \frac{1}{k} \geq n \cdot \frac{1}{2n} = \frac{1}{2}$ for every n, so no N works for $\varepsilon = \frac{1}{2}$. Note that $x_{n+1} - x_n \to 0$ is *not* enough for a sequence to be Cauchy. □

Exercise 4.3. [Medium] Show that $f(x) = \frac{1}{2}(x + \frac{2}{x})$ is a contraction of $[1,2]$ and find its fixed point.

Solution. $f'(x) = \frac{1}{2}(1 - \frac{2}{x^2}) \in [-\frac{1}{2}, \frac{1}{4}]$ on $[1,2]$, so $|f'| \leq \frac{1}{2}$ and f is a contraction with $k = \frac{1}{2}$. Also f has minimum $f(\sqrt{2}) = \sqrt{2}$ and $f(1) = f(2) = \frac{3}{2}$, so $f([1,2]) \subseteq [\sqrt{2}, \frac{3}{2}] \subseteq [1,2]$. The fixed point solves $x = \frac{1}{2}(x + \frac{2}{x})$, so $x^2 = 2$ and $x^* = \sqrt{2}$. On $[1,2] \cap \mathbb{Q}$, f is still a contraction of the space into itself but has no fixed point, which shows completeness is needed. □

Exercise 4.4. [Hard] Show that $(C[0,1], d_1)$ is not complete.

Solution. For $n \geq 2$ let

$$f_n(t) = \begin{cases} 0, & 0 \leq t \leq \frac{1}{2}, \\ n(t - \frac{1}{2}), & \frac{1}{2} \leq t \leq \frac{1}{2} + \frac{1}{n}, \\ 1, & \frac{1}{2} + \frac{1}{n} \leq t \leq 1. \end{cases}$$

For $m > n$, f_m and f_n differ only on $[\frac{1}{2}, \frac{1}{2} + \frac{1}{n}]$, and by at most 1, so $d_1(f_m, f_n) \leq \frac{1}{n}$. The sequence is Cauchy.

Suppose $d_1(f_n, f) \to 0$ with $f \in C[0,1]$. Then $\int_0^{1/2} |f| = \int_0^{1/2} |f - f_n| \leq d_1(f, f_n) \to 0$, so $f = 0$ on $[0, \frac{1}{2}]$ by continuity. For any $c > \frac{1}{2}$ and all n with $\frac{1}{2} + \frac{1}{n} < c$, $\int_c^1 |f - 1| \leq d_1(f, f_n) \to 0$, so $f = 1$ on $[c, 1]$. Hence $f = 1$ on $(\frac{1}{2}, 1]$. Then f is discontinuous at $\frac{1}{2}$, a contradiction. □

Exercise 4.5. [Hard] Let X be complete and $f\colon X \to X$ such that the iterate f^m is a contraction for some $m \geq 1$. Prove that f has a unique fixed point.

Solution. By Theorem 4.9, f^m has a unique fixed point x^*. Then $f^m(f(x^*)) = f(f^m(x^*)) = f(x^*)$, so $f(x^*)$ is also a fixed point of f^m. By uniqueness, $f(x^*) = x^*$. Any fixed point of f is a fixed point of f^m, so it is unique. □

Chapter 5

Compactness

Study slot: 5:10–5:35 (25 min)

Learn the open-cover definition word for word, the three equivalent forms for metric spaces (Theorem 5.3), Heine–Borel and where it fails, and the theorems about continuous maps on compact spaces. Practise proving that a set is *not* compact by giving a bad open cover or a bad sequence.

Intuition. Compactness is the topological version of "finite". Finite sets have convenient properties: every real function on them has a maximum, and every sequence in them repeats some value infinitely often. Compact sets keep these properties even when infinite. In $\mathbb{R}^n$, compact simply means *closed and bounded*.

5.1 Definitions

Definition 5.1 (Open cover, compactness)**.** Let $K \subseteq X$.

(i) An **open cover** of K is a family $\mathcal{U} = \{U_i\}_{i \in I}$ of open sets with $K \subseteq \bigcup_{i \in I} U_i$.

(ii) A **subcover** is a subfamily that still covers K. It is **finite** if it has finitely many members.

(iii) K is **compact** if *every* open cover of K has a finite subcover:

$$K \subseteq \bigcup_{i \in I} U_i \implies K \subseteq U_{i_1} \cup \cdots \cup U_{i_n} \quad \text{for some } i_1, \ldots, i_n \in I.$$

Warning (The most common misreading). "K has a finite open cover" is always true, since $\{X\}$ is one. The definition requires that *every* open cover can be reduced to finitely many sets. To prove compactness, handle an *arbitrary* cover. To disprove it, exhibit *one* cover with no finite subcover.

Definition 5.2 (Sequential compactness, total boundedness)**.**

(i) X is **sequentially compact** if every sequence in X has a subsequence converging to a point *of* X.

(ii) X is **totally bounded** if for every $\varepsilon > 0$ there are finitely many points $x_1, \ldots, x_n$ (an **ε-net**) with

$$X = B(x_1, \varepsilon) \cup \cdots \cup B(x_n, \varepsilon).$$

Totally bounded $\Rightarrow$ bounded, but not conversely. An infinite discrete space has diam $= 1$, but with $\varepsilon = \frac{1}{2}$ each ball is a single point.

5.2 The main equivalence

Theorem 5.3. *For a metric space X, the following are equivalent:*

(1) X is compact;

(2) X is sequentially compact;

(3) X is complete and totally bounded.

Sketch of proof. (1)⇒(2): Suppose (x_n) has no convergent subsequence. Then each $x \in X$ has a ball $B(x, r_x)$ containing x_n for only finitely many n. Finitely many of these balls cover X, so only finitely many indices n exist, a contradiction.

(2)⇒(3): A Cauchy sequence has a convergent subsequence, so it converges (Proposition 4.2). If X were not totally bounded for some ε, we could inductively choose $x_{n+1} \notin \bigcup_{k \le n} B(x_k, \varepsilon)$. Then $d(x_m, x_n) \ge \varepsilon$ for all $m \neq n$, so no subsequence is Cauchy.

(3)⇒(2): Cover X by finitely many balls of radius 1. One contains infinitely many terms. Repeat inside that with radius $\frac{1}{2}, \frac{1}{3}, \dots$ and take a diagonal subsequence. It is Cauchy, so it converges by completeness.

(2)⇒(1): Use the Lebesgue number δ of the cover (Lemma 5.11) and a finite $\frac{\delta}{3}$-net. Each ball of the net lies inside a single member of the cover. □

5.3 Heine–Borel and its limits

Theorem 5.4 (Compact ⇒ closed and bounded)**.** *A compact subset K of a metric space X is closed and bounded.*

Proof. Bounded: $\{B(x_0, n)\}_{n \in \mathbb{N}}$ is an open cover of K, and a finite subcover gives $K \subseteq B(x_0, N)$. *Closed:* let $k_n \in K$ with $k_n \to x$. Some subsequence converges to a point of K. That subsequence also converges to x, so by uniqueness of limits $x \in K$. □

Theorem 5.5 (Heine–Borel)**.** *A subset of $(\mathbb{R}^n, d_2)$ is compact if and only if it is closed and bounded.*

Proof. (⇒) Theorem 5.4. (⇐) A sequence in a bounded set has a convergent subsequence by Bolzano–Weierstrass, applied one coordinate at a time. Its limit lies in the set because the set is closed. So the set is sequentially compact. □

Warning (Closed and bounded ⇏ compact in general).

(a) An infinite discrete space is closed and bounded, but the cover $\{\{x\}\}_{x \in X}$ has no finite subcover.

(b) In the closed unit ball of ℓ^2, the vectors $e_n = (0, \dots, 0, 1, 0, \dots)$ satisfy $d(e_n, e_m) = \sqrt{2}$ for $n \neq m$, so no subsequence is Cauchy.

(c) In the closed unit ball of $(C[0,1], d_\infty)$, $f_n(t) = t^n$ has no uniformly convergent subsequence, because its pointwise limit is discontinuous.

(d) $X = (0, 1]$ is closed in itself and bounded, but $(\frac{1}{n})$ has no subsequence converging in X.

In general, the correct replacement for "closed and bounded" is *complete and totally bounded*.

5.4 Key theorems

Theorem 5.6. *A closed subset of a compact space is compact.*

Proof. Let $F \subseteq K$ be closed and $\{U_i\}$ an open cover of F. Then $\{U_i\} \cup \{X \setminus F\}$ is an open cover of K. Take a finite subcover and discard $X \setminus F$; the remaining sets cover F. □

Theorem 5.7 (Continuous image of a compact set)**.** *If $f\colon X \to Y$ is continuous and $K \subseteq X$ is compact, then $f(K)$ is compact.*

Proof. Let $\{V_i\}$ be an open cover of $f(K)$. Each $f^{-1}(V_i)$ is open (Theorem 3.2), and together they cover K. Choose a finite subcover $f^{-1}(V_1), \dots, f^{-1}(V_n)$. Then $f(K) \subseteq V_1 \cup \cdots \cup V_n$. □

Corollary 5.8 (Extreme Value Theorem)**.** *If $K \neq \emptyset$ is compact and $f\colon K \to \mathbb{R}$ is continuous, then f is bounded and attains its bounds: there exist $a, b \in K$ with*

$$f(a) = \inf_{x \in K} f(x), \qquad f(b) = \sup_{x \in K} f(x).$$

Proof. $f(K)$ is compact, hence closed and bounded in $\mathbb{R}$, so it contains its supremum and infimum. □

Without compactness this fails: $f(x) = x$ on $(0, 1)$ attains neither bound, and $1/x$ on $(0, 1]$ is unbounded.

Theorem 5.9 (Heine–Cantor)**.** *A continuous map from a compact metric space to a metric space is uniformly continuous.*

Proof. If not, by Proposition 3.7 there are $\varepsilon > 0$ and x_n, y_n with $d(x_n, y_n) < \frac{1}{n}$ but $d\big(f(x_n), f(y_n)\big) \geq \varepsilon$. By sequential compactness, $x_{n_k} \to x$ for some subsequence. Then $y_{n_k} \to x$ as well, since $d(y_{n_k}, x) \leq \frac{1}{n_k} + d(x_{n_k}, x)$. By continuity, $f(x_{n_k})$ and $f(y_{n_k})$ both converge to $f(x)$, so their distance tends to 0, a contradiction. □

Theorem 5.10. *Let $f\colon X \to Y$ be a continuous bijection, with X compact and Y Hausdorff (for example, a metric space). Then f is a homeomorphism.*

Proof. We must show f^{-1} is continuous, that is, $(f^{-1})^{-1}(C) = f(C)$ is closed for each closed $C \subseteq X$. C is compact (Theorem 5.6), so $f(C)$ is compact (Theorem 5.7), and compact subsets of Hausdorff spaces are closed. □

Lemma 5.11 (Lebesgue Number Lemma)**.** *If X is a compact metric space and $\mathcal{U}$ is an open cover of X, there is $\delta > 0$ (a **Lebesgue number**) such that every subset of X of diameter less than δ lies in a single member of $\mathcal{U}$.*

Theorem 5.12 (Finite intersection property)**.** *A family of sets has the **finite intersection property** (FIP) if every finite subfamily has non-empty intersection. X is compact $\iff$ every family of closed sets with the FIP has non-empty intersection. In particular, nested non-empty compact sets $K_1 \supseteq K_2 \supseteq \cdots$ satisfy $\bigcap_n K_n \neq \emptyset$.*

Proposition 5.13 (Further facts)**.**

(i) Finite unions of compact sets are compact.

(ii) A compact metric space is complete, totally bounded, bounded and separable.

(iii) $X \times Y$ *is compact* $\iff$ X *and* Y *are compact (Tychonoff's theorem extends this to arbitrary products).*

(iv) If K *is compact,* F *is closed and* $K \cap F = \emptyset$*, then* $d(K, F) > 0$*. This fails for two closed sets: the* x*-axis and* $\{\,(x, y) \,:\, y = 1/x,\ x > 0\,\}$ *are disjoint but at distance* 0*.*

(v) Compactness is a topological property.

Example 5.14 (Compactness in topological spaces)**.** Finite spaces, indiscrete spaces and cofinite spaces are compact. For a cofinite space, one member of any open cover misses only finitely many points, and finitely many more members cover those. An infinite discrete space is not compact.

5.5 Exercises

Exercise 5.1. [Easy] Show directly from the definition that $(0, 1)$ is not compact.

Solution. The sets $U_n = (\frac{1}{n}, 1)$, $n \geq 2$, cover $(0, 1)$. A finite subfamily has union $(\frac{1}{N}, 1)$, where N is its largest index, and this misses $\frac{1}{2N}$. □

Exercise 5.2. [Easy] Which of these are compact? (a) $[0, 1] \cup [2, 3]$; (b) $\{0\} \cup \left\{\, \frac{1}{n} \,:\, n \in \mathbb{N} \,\right\}$; (c) $\mathbb{Z}$; (d) $S^1 = \{\,(x, y) \,:\, x^2 + y^2 = 1\,\}$; (e) $\mathbb{Q} \cap [0, 1]$.

Solution. (a) Yes: closed and bounded. (b) Yes: closed, since it contains its only limit point 0, and bounded. (c) No: unbounded. (d) Yes: closed (a preimage of $\{1\}$) and bounded. (e) No: not closed, since $\frac{\sqrt{2}}{2}$ is a limit of points in the set but not in it. □

Exercise 5.3. [Medium] Prove from the open-cover definition that $K = \{0\} \cup \left\{\, \frac{1}{n} \,:\, n \in \mathbb{N} \,\right\}$ is compact.

Solution. Let $\mathcal{U}$ be an open cover. Some $U_0 \in \mathcal{U}$ contains 0, and hence contains $(-\varepsilon, \varepsilon)$ for some $\varepsilon > 0$, so it contains $\frac{1}{n}$ for all $n > \frac{1}{\varepsilon}$. The finitely many remaining points $1, \frac{1}{2}, \ldots, \frac{1}{N}$ each lie in some member of $\mathcal{U}$. That gives at most $N + 1$ sets in total. The limit point swallows the tail. □

Exercise 5.4. [Medium] Let K be compact and $f \colon K \to \mathbb{R}$ continuous with $f > 0$. Show $\inf_K f > 0$.

Solution. By Corollary 5.8, $\inf_K f = f(x_0)$ for some $x_0 \in K$, and $f(x_0) > 0$. Compare $f(x) = x$ on $(0, 1]$, where the infimum 0 is not attained. □

Exercise 5.5. [Hard] Let (X, d) be compact and $f \colon X \to X$ satisfy $d(f(x), f(y)) < d(x, y)$ for all $x \neq y$. Prove that f has a unique fixed point. Compare with $x + \frac{1}{x}$ on $[1, \infty)$ in Chapter 4.

Solution. $g(x) = d(x, f(x))$ is continuous on the compact space X, so it attains its minimum at some x_0. If $f(x_0) \neq x_0$, then

$$g(f(x_0)) = d(f(x_0), f(f(x_0))) < d(x_0, f(x_0)) = g(x_0),$$

contradicting minimality. So $f(x_0) = x_0$. Uniqueness: two distinct fixed points $x \neq y$ would give $d(x, y) < d(x, y)$. Compactness replaces the uniform constant $k < 1$. □

Exercise 5.6. [Hard] Show that an isometry $f\colon X \to X$ of a compact metric space is surjective.

Solution. Suppose $a \notin f(X)$. $f(X)$ is compact, hence closed, so $\varepsilon = d(a, f(X)) > 0$. Let $x_n = f^n(a)$. For $m > n$, since f^n preserves distances,

$$d(x_n, x_m) = d\big(a, f^{m-n}(a)\big) \geq \varepsilon,$$

because $f^{m-n}(a) \in f(X)$. So (x_n) has no Cauchy subsequence, contradicting sequential compactness. □

Appendix A

Connectedness

Some courses put connectedness in Chapter 5 or 6. It is short and often appears in questions about homeomorphisms.

Definition A.1.

(i) X is **disconnected** if $X = U \cup V$ with U, V open, disjoint and non-empty (a **separation**). Otherwise X is **connected**. Equivalently, the only clopen subsets are $\emptyset$ and X, or every continuous $f\colon X \to \{0,1\}$ is constant.

(ii) X is **path-connected** if for all $x, y \in X$ there is a continuous $\gamma\colon [0,1] \to X$ with $\gamma(0) = x$ and $\gamma(1) = y$.

(iii) A **component** is a maximal connected subset.

Theorem A.2.

(i) The connected subsets of $\mathbb{R}$ are exactly the intervals.

*(ii) Continuous images of connected spaces are connected. This gives the **Intermediate Value Theorem**.*

*(iii) Path-connected $\Rightarrow$ connected. The converse fails for the **topologist's sine curve***

$$S = \left\{ \left(x, \sin \tfrac{1}{x}\right) \;:\; x > 0 \right\} \cup (\{0\} \times [-1,1]).$$

(iv) Open connected subsets of $\mathbb{R}^n$ are path-connected.

(v) The closure of a connected set is connected. A union of connected sets with a common point is connected.

(vi) $\mathbb{Q}$ is totally disconnected: $\left\{ q \;:\; q < \sqrt{2} \right\}$ and $\left\{ q \;:\; q > \sqrt{2} \right\}$ separate it.

Exam tip (Using connectedness to show $X \ncong Y$). If $f\colon X \to Y$ is a homeomorphism, then $X \setminus \{p\} \cong Y \setminus \{f(p)\}$. Removing any point disconnects $\mathbb{R}$ but not $\mathbb{R}^2$, so $\mathbb{R} \ncong \mathbb{R}^2$. Similarly $[0,1) \ncong (0,1)$ and $S^1 \ncong [0,1]$.

Appendix B

Proof-Template Toolkit

Most exam questions follow one of these patterns. Learn the first move for each.

To prove...	**Standard first move**
d is a metric	Check (M1), (M2), (M3). For (M3), add and subtract, or use an increasing subadditive φ.
d is not a metric	One numerical counterexample to one axiom.
U is open	"Let $x \in U$. Take $r = \dots$", then show $B(x, r) \subseteq U$. Or write $U = f^{-1}(\text{open})$ with f continuous.
F is closed	Show the complement is open; or $F = f^{-1}(\text{closed})$; or $x_n \in F,\ x_n \to x \Rightarrow x \in F$.
A set is not open	One point of the set where every ball leaves the set.
A set is not closed	A sequence in the set converging to a point outside it.
$x \in \operatorname{cl} A$	A sequence in A converging to x, or show every ball around x meets A.
f is continuous	ε–δ; build it from known continuous maps; or show it is Lipschitz.
f is not continuous	One sequence $x_n \to x$ with $f(x_n) \not\to f(x)$.
f is not uniformly continuous	$d(x_n, y_n) \to 0$ but $d\big(f(x_n), f(y_n)\big) \geq \varepsilon$.
$X \cong Y$	An explicit bijection, continuous in both directions; or a continuous bijection from a compact space to a Hausdorff space.
$X \ncong Y$	A topological property that differs: compactness, connectedness, components after removing a point, Hausdorff, separability.
(x_n) is Cauchy	Bound $d(x_m, x_n)$ for $m > n$ by a quantity depending only on n that tends to 0. A geometric series often works.
X is complete	A closed subset of a complete space, or the 3-step template: candidate limit, convergence in the metric, membership.
X is not complete	One Cauchy sequence whose limit is missing.
A unique solution exists	The Contraction Mapping Theorem: complete space, $f(X) \subseteq X$, constant $k < 1$.
K is compact	In $\mathbb{R}^n$: closed and bounded. Otherwise: sequential compactness, a closed subset of a compact set, a continuous image of a compact set, or an arbitrary open cover.
K is not compact	One open cover with no finite subcover; one sequence with no convergent subsequence; or show it is not closed, or not bounded.
(X, τ) is not metrizable	Show it is not Hausdorff.

Appendix C

Counterexample Bank

False statement	**Counterexample**
Not open $\Rightarrow$ closed	$[0,1)$, or $\mathbb{Q}$, in $\mathbb{R}$
Infinite intersections of open sets are open	$\bigcap_n (-\frac{1}{n}, \frac{1}{n}) = \{0\}$
Infinite unions of closed sets are closed	$\bigcup_n [\frac{1}{n}, 1] = (0,1]$
$\operatorname{cl} B(x,r) = \overline{B}(x,r)$	discrete metric with $r = 1$
$\operatorname{cl}(A \cap B) = \operatorname{cl} A \cap \operatorname{cl} B$	$A = \mathbb{Q}$, $B = \mathbb{R} \setminus \mathbb{Q}$
Limits are unique in every topological space	indiscrete or cofinite topology
Continuous images of open or closed sets are open or closed	x^2 on $(-1,1)$; arctan on $\mathbb{R}$
Continuous $\Rightarrow$ uniformly continuous	x^2 on $\mathbb{R}$; $1/x$ on $(0,1)$
Uniformly continuous $\Rightarrow$ Lipschitz	$\sqrt{x}$ on $[0,1]$
A continuous bijection is a homeomorphism	$[0, 2\pi) \to S^1$; $\mathrm{id}\colon (\mathbb{R}, \mathrm{disc}) \to (\mathbb{R}, \lvert\cdot\rvert)$
Boundedness and completeness are topological	$\mathbb{R} \cong (0,1)$
Topologically equivalent $\Rightarrow$ Lipschitz equivalent	d and $d/(1+d)$ on $\mathbb{R}$
d_1 and d_∞ on $C[0,1]$ are equivalent	t^n: $d_1 \to 0$ but $d_\infty = 1$
$x_{n+1} - x_n \to 0 \Rightarrow$ Cauchy	$\sum_{k \le n} \frac{1}{k}$; $\sqrt{n}$
Every Cauchy sequence converges	$\frac{1}{n}$ in $(0,1)$; approximations to $\sqrt{2}$ in $\mathbb{Q}$
$d(fx, fy) < d(x,y) \Rightarrow$ fixed point	$x + \frac{1}{x}$ on $[1, \infty)$ (true if X is compact)
A contraction on any space has a fixed point	$x/2$ on $(0,1]$
Nested non-empty closed sets intersect	$[n, \infty)$ in $\mathbb{R}$
Closed and bounded $\Rightarrow$ compact	infinite discrete space; unit ball of ℓ^2; $(0,1]$ in itself
Bounded $\Rightarrow$ totally bounded	infinite discrete space
Disjoint closed sets are a positive distance apart	x-axis and $y = 1/x$
Connected $\Rightarrow$ path-connected	topologist's sine curve

Appendix D

Glossary

Accumulation / limit point of A
Every ball around x meets $A \setminus \{x\}$.

Banach space
A complete normed vector space.

Basis for a topology
A family of open sets such that every open set is a union of its members.

Boundary ∂A
$\operatorname{cl} A \setminus \operatorname{int} A$: every ball around the point meets both A and $X \setminus A$.

Bounded
Contained in some ball; equivalently, $\operatorname{diam} A < \infty$.

Cauchy sequence
$\forall \varepsilon \; \exists N \; \forall m, n \geq N : \; d(x_m, x_n) < \varepsilon$.

Clopen
Both open and closed.

Closed set
Its complement is open; equivalently, it contains the limits of its convergent sequences.

Closure $\operatorname{cl} A$
The smallest closed superset of A; equals $A \cup A'$.

Cofinite topology
$\emptyset$ together with the sets that have finite complement.

Compact
Every open cover has a finite subcover.

Complete
Every Cauchy sequence converges in the space.

Completion
A complete space containing X isometrically as a dense subset.

Connected
Not a union of two disjoint non-empty open sets.

Continuous
Preimages of open sets are open; equivalently, the ε–δ or sequential conditions.

Contraction
$d(f(x), f(y)) \leq k\, d(x, y)$ for a fixed $k < 1$.

Dense
$\operatorname{cl} A = X$; equivalently, A meets every non-empty open set.

Diameter
$\operatorname{diam} A = \sup \{\, d(a, b) \;:\; a, b \in A \,\}$.

Discrete metric
$d(x, y) = 1$ for $x \neq y$; every set is open.

Distance to a set
$d(x, A) = \inf_{a \in A} d(x, a)$; it is $0 \iff x \in \operatorname{cl} A$.

Equivalent metrics
Lipschitz: $\alpha d \le d' \le \beta d$. Topological: the same open sets.

Finite intersection property
Every finite subfamily has non-empty intersection.

Fixed point
A point with $f(x) = x$.

Hausdorff
Distinct points have disjoint open neighbourhoods.

Homeomorphism
A continuous bijection with a continuous inverse.

Interior $\operatorname{int} A$
The largest open subset of A.

Isolated point
A point $a \in A$ with $B(a, r) \cap A = \{a\}$ for some $r > 0$.

Isometry
A distance-preserving map.

Lebesgue number
A δ such that every set of diameter $< \delta$ lies in one member of the cover.

Lipschitz
$d(f(x), f(y)) \le L\, d(x, y)$.

Metric
Satisfies (M1) positivity, (M2) symmetry and (M3) the triangle inequality.

Metrizable
The topology comes from a metric.

Norm
Satisfies (N1)–(N3); induces the metric $\|u - v\|$.

Nowhere dense
$\operatorname{int}(\operatorname{cl} A) = \emptyset$.

Open cover
A family of open sets whose union contains the set.

Open set
Every point has a ball around it inside the set.

Path-connected
Any two points are joined by a continuous path.

Separable
Has a countable dense subset.

Sequentially compact
Every sequence has a subsequence converging in the space.

Topological property
A property preserved by homeomorphisms.

Topology
Contains $\emptyset$ and X; closed under arbitrary unions and finite intersections.

Totally bounded
For every $\varepsilon > 0$, finitely many ε-balls cover the space.

Uniformly continuous
The ε–δ condition with δ independent of the point.

Appendix E

Recap Checklist (5:35–5:45)

Close the notes. On a blank page, write as many of these as you can from memory. Tick off what you got and circle what you missed. The circled items are where to start next session.

Definitions to state exactly

- ☐ Metric (M1)–(M3); norm
- ☐ Open set, closed set; interior, closure, boundary, limit point, dense
- ☐ Convergent sequence; topology (T1)–(T3); Hausdorff
- ☐ Continuity (ε–δ and open-set versions); uniform continuity; Lipschitz
- ☐ Homeomorphism; topological property
- ☐ Cauchy sequence; complete; contraction
- ☐ Compact (open covers); sequentially compact; totally bounded

Theorems to state, with a proof sketch

- ☐ Open balls are open (the radius $r - d(x, y)$)
- ☐ Unions and finite intersections of open sets are open, with a counterexample for infinite intersections
- ☐ The characterisations of the closure
- ☐ The four equivalent forms of continuity
- ☐ Closed subsets of complete spaces are complete; complete subspaces are closed
- ☐ $(C[a, b], d_\infty)$ is complete (the $\varepsilon/3$ argument)
- ☐ Contraction Mapping Theorem: all five steps
- ☐ Compact $\iff$ sequentially compact $\iff$ complete and totally bounded
- ☐ Heine–Borel, and a counterexample outside $\mathbb{R}^n$
- ☐ The continuous image of a compact set is compact, hence the Extreme Value Theorem
- ☐ Heine–Cantor; a continuous bijection from a compact space to a Hausdorff space is a homeomorphism

Counterexamples to recall instantly

- ☐ A set that is neither open nor closed
- ☐ A continuous function that is not uniformly continuous
- ☐ A continuous bijection that is not a homeomorphism

☐ An incomplete space, with its Cauchy sequence

☐ A closed, bounded, non-compact set

> *Exam tip* (Next up). Break 5:45–6:00, then **linear systems 6:00–6:50** (separate notes), then wrap-up at 6:50 and a hard stop at **7:00 pm**.